

Tutorial: Introduction to Game Theory (1/2)

IEDA 1250 Optimizing Decisions for Personal and Business Development

Wu Hongfan

The Hong Kong University of Science and Technology

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Plan for Today

- 1 Translate a story into players, strategies, and payoffs.
- 2 Find Nash equilibria by best responses and dominated strategies.
- 3 Solve zero-sum games with saddle points and mixed strategies.
- 4 Connect the method to homework-style questions.

Target

By the end, you should be able to read a payoff table and decide which solution method to try first.

What Makes a Situation a Game?

A game

A decision situation where each player's payoff depends on the actions of more than one decision maker.

Ingredients

- Players
- Strategies
- Payoffs
- Information and timing

Typical assumptions

- Rationality
- Self-interest
- Common knowledge of the game
- Simultaneous or sequential moves

Example: Prisoner's Dilemma

		Prisoner B	
		Silent	Betray
Prisoner A	Silent	-2, -2	-10, 0
	Betray	0, -10	-5, -5

- Payoffs are utilities, so fewer years in prison means a larger payoff.
- Each player chooses without observing the other player's choice.
- The social outcome (Silent, Silent) is better for both, but it is not stable.

Best Response Thinking

Best response

A strategy is a best response if it gives the highest payoff given the other player's strategy.

		Prisoner B	
		Silent	Betray
Prisoner A	Silent	-2, -2	-10, 0
	Betray	0, -10	-5, -5

- If B is silent, A prefers Betray: $0 > -2$.
- If B betrays, A prefers Betray: $-5 > -10$.
- By symmetry, B also prefers Betray in both cases.

Nash Equilibrium

Nash equilibrium

A strategy profile where no single player can improve by changing only their own strategy.

- “I am doing my best given what you do.”
- “You are doing your best given what I do.”
- It is a stability concept, not necessarily a fairness or efficiency concept.

Prisoner's Dilemma

The Nash equilibrium is (Betray, Betray).

How to Read a Payoff Matrix

		Column player	
		L	R
Row player	U	3, 2	0, 3
	D	2, 1	1, 0

- First number: row player's payoff.
- Second number: column player's payoff.
- Row player chooses a row.
- Column player chooses a column.

Check

Never compare payoffs across different players. Compare a player's own payoffs while holding the other player's action fixed.

Dominated Strategies

Dominated strategy

Strategy s is dominated if another strategy is always at least as good and sometimes strictly better, regardless of the opponent's action.

- For a row player who maximizes: larger row payoffs are better.
- For a column player in a zero-sum payoff table for Player 1: smaller entries are better.
- Eliminating dominated strategies can simplify a game before solving it.

Exercise 1: Eliminate Dominated Strategies

Player 1	Player 2				
	S1	S2	S3	S4	S5
S1	3	2	0	3	0
S2	5	0	2	5	9
S3	7	3	9	5	9
S4	4	6	8	7	5.5
S5	6	0	8	8	3

Task

Assume the entries are payoffs to Player 1. Which strategies can we remove first?

Exercise 1: One Elimination Order

- 1 Column S4 is dominated by Column S2, so remove Column S4.
- 2 Row S1 is dominated by Row S3, so remove Row S1.
- 3 Row S2 is dominated by Row S3, so remove Row S2.
- 4 In the reduced table, Row S5 is dominated by Row S3, so remove Row S5.
- 5 In the reduced table, Column S3 is dominated by Column S5, so remove Column S3.

Takeaway

Dominance is a simplification tool. The order may differ, but every removal must be justified by comparing all remaining entries.

Zero-Sum Games

Zero-sum game

One player's gain is exactly the other player's loss.

- A single payoff table is enough: entries are payoffs to Player A.
- Player A wants to maximize the payoff.
- Player B wants to minimize Player A's payoff.

Value of the game

The expected payoff to Player A when both players use optimal strategies.

Pure Strategy: Saddle Point Test

	B1	B2	B3	B4	Row min
A1	20	15	12	35	12
A2	25	14	8	10	8
A3	40	2	10	5	2
A4	-5	4	11	0	-5
Col max	40	15	12	35	

- Player A chooses the row with largest row minimum: $\max \min = 12$.
- Player B chooses the column with smallest column maximum: $\min \max = 12$.
- Since they are equal, there is a saddle point.

Pure Strategy: Result

Saddle point

The saddle point is $(A1, B3)$ and the value of the game is 12.

- It is the minimum in Row A1.
- It is the maximum in Column B3.
- If A announces A1, B cannot reduce A below 12.
- If B announces B3, A cannot increase above 12.

When There Is No Saddle Point

	B1	B2	B3	Row min
A1	0	-2	2	-2
A2	5	4	-3	-3
A3	2	3	-4	-4
Col max	5	4	2	

- $\max \min = -2$.
- $\min \max = 2$.
- They are not equal, so no pure strategy saddle point exists.
- We need mixed strategies.

Mixed Strategies

Mixed strategy

A probability distribution over pure strategies.

- Player A may choose A1 with probability p and A2 with probability $1 - p$.
- Player B may choose B1 with probability q and B2 with probability $1 - q$.
- The payoff becomes an expected payoff.

Key idea

At equilibrium, a player randomizes only among strategies that give the same expected payoff.

Graphical Method: Setup

Use the problem set example:

	B1	B2	B3	B4
A1	2	2	3	-2
A2	4	3	2	6

Let $p = \Pr(A1)$ and $1 - p = \Pr(A2)$. Expected payoffs to A:

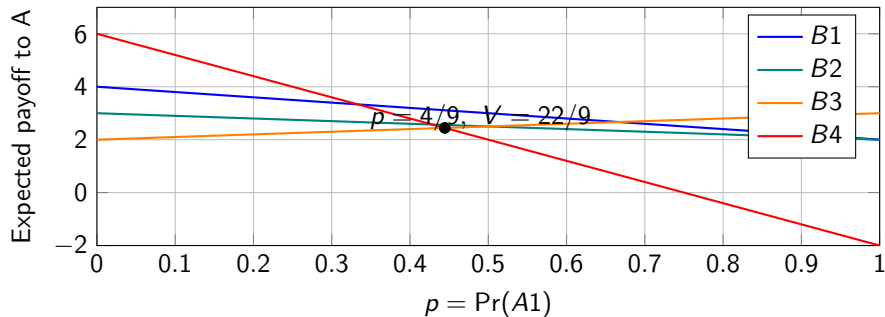
$$f_{B1}(p) = 2p + 4(1 - p) = 4 - 2p,$$

$$f_{B2}(p) = 2p + 3(1 - p) = 3 - p,$$

$$f_{B3}(p) = 3p + 2(1 - p) = 2 + p,$$

$$f_{B4}(p) = -2p + 6(1 - p) = 6 - 8p.$$

Graphical Method: Maximize the Lower Envelope



Result for Player A

Choose $(A1, A2) = \left(\frac{4}{9}, \frac{5}{9}\right)$ and guarantee value $V = \frac{22}{9}$.

Why the Intersection Matters

- Player B chooses the column that gives A the smallest expected payoff.
- So A cares about the minimum of the four lines.
- A chooses p to make this minimum as large as possible.
- The best point occurs where the binding lines $B3$ and $B4$ meet:

$$2 + p = 6 - 8p \quad \Rightarrow \quad p = \frac{4}{9}.$$

$$V = 2 + \frac{4}{9} = \frac{22}{9}.$$

Check the Binding Lines

At $p = 4/9$:

$$f_{B1}(4/9) = 4 - 8/9 = 28/9,$$

$$f_{B2}(4/9) = 3 - 4/9 = 23/9,$$

$$f_{B3}(4/9) = 2 + 4/9 = 22/9,$$

$$f_{B4}(4/9) = 6 - 32/9 = 22/9.$$

Binding constraints

The binding columns are $B3$ and $B4$, so Player B places positive probability only on $B3$ and $B4$.

Find Player B's Mixed Strategy

Let $q_j = \Pr(B_j)$. Since $B3$ and $B4$ are binding, try

$$q_1 = q_2 = 0, \quad q_3 + q_4 = 1.$$

Make A indifferent between $A1$ and $A2$:

$$3q_3 - 2q_4 = 2q_3 + 6q_4.$$

$$q_3 = 8q_4, \quad q_3 + q_4 = 1.$$

Result for Player B

$$(B1, B2, B3, B4) = \left(0, 0, \frac{8}{9}, \frac{1}{9}\right).$$

Graphical Method Checklist

- 1 If Player A has two rows, let p be the probability of the first row.
- 2 Write one expected payoff line for each column of Player B.
- 3 Draw or compare the lower envelope.
- 4 Choose the p that maximizes the lower envelope.
- 5 Binding columns get positive probability for Player B.
- 6 Use indifference equations to find Player B's probabilities.

Practice 2: A 6-by-2 Graphical Problem

	B1	B2
A1	1	-3
A2	3	5
A3	-1	6
A4	4	1
A5	2	2
A6	-5	0

Task

This time Player B has two columns. Let $q = \Pr(B1)$. Which row payoff lines should determine the solution?

Practice 2: Payoff Lines

$$g_{A1}(q) = 1q - 3(1 - q) = -3 + 4q,$$

$$g_{A2}(q) = 3q + 5(1 - q) = 5 - 2q,$$

$$g_{A3}(q) = -q + 6(1 - q) = 6 - 7q,$$

$$g_{A4}(q) = 4q + 1(1 - q) = 1 + 3q,$$

$$g_{A5}(q) = 2,$$

$$g_{A6}(q) = -5q.$$

Player B's goal

Player B chooses q to minimize the upper envelope, because A will choose the best row after seeing expected payoffs.

Practice 2: Result

The solution is determined by rows A2 and A4:

$$5 - 2q = 1 + 3q \Rightarrow q = \frac{4}{5}.$$

$$V = 5 - 2 \cdot \frac{4}{5} = \frac{17}{5} = 3.4.$$

To find A's probabilities, let A mix A2 with probability p and A4 with probability $1 - p$:

$$3p + 4(1 - p) = 5p + 1(1 - p).$$

$$4 - p = 1 + 4p \Rightarrow p = \frac{3}{5}.$$

Optimal strategies

A: $(0, \frac{3}{5}, 0, \frac{2}{5}, 0, 0)$, B: $(\frac{4}{5}, \frac{1}{5})$.

LP Formulation for Zero-Sum Games

For Player A with mixed strategy x_i and game value V :

$$\begin{aligned} & \max V \\ & \sum_i a_{ij}x_i \geq V \quad \text{for every column } j \\ & \sum_i x_i = 1, \quad x_i \geq 0. \end{aligned}$$

- Each column constraint says: if B chooses this column, A's expected payoff is at least V .
- Maximize the guaranteed payoff.
- This is another way to express the minimax idea.

From Zero-Sum to Business Competition

- Not all games are zero-sum.
- In business, firms may both earn positive profits.
- The Nash equilibrium idea still applies: each firm chooses a best response to the other.

Cournot competition

Firms choose quantities simultaneously. The market price depends on total quantity.

Cournot Model

Two firms choose q_1 and q_2 .

$$p = a - b(q_1 + q_2)$$

If both firms have unit cost c , firm 1's profit is

$$\pi_1 = [a - b(q_1 + q_2) - c]q_1.$$

Firm 1's best response:

$$q_1^* = \frac{a - c - bq_2}{2b}.$$

Similarly,

$$q_2^* = \frac{a - c - bq_1}{2b}.$$

Cournot Equilibrium

Solve the two best-response equations:

$$q_1^* = \frac{a - c - bq_2}{2b}, \quad q_2^* = \frac{a - c - bq_1}{2b}.$$

For symmetric firms:

$$q_1^* = q_2^* = \frac{a - c}{3b}.$$

Interpretation

Each firm produces less than the perfectly competitive level, because each firm knows that more output lowers the market price.

Cournot Practice

Problem

Market price is $p = 14 - 2Q$, where $Q = q_1 + q_2$. Firm 1 has unit cost 2 and Firm 2 has unit cost 4. Find the equilibrium quantities.

$$\pi_1 = (14 - 2q_1 - 2q_2 - 2)q_1 = (12 - 2q_1 - 2q_2)q_1.$$

$$\pi_2 = (14 - 2q_1 - 2q_2 - 4)q_2 = (10 - 2q_1 - 2q_2)q_2.$$

Cournot Practice: Solution

First-order conditions:

$$\frac{\partial \pi_1}{\partial q_1} = 12 - 4q_1 - 2q_2 = 0 \Rightarrow q_1 = 3 - \frac{1}{2}q_2.$$

$$\frac{\partial \pi_2}{\partial q_2} = 10 - 2q_1 - 4q_2 = 0 \Rightarrow q_2 = 2.5 - \frac{1}{2}q_1.$$

Solve:

$$q_1^* = \frac{7}{3}, \quad q_2^* = \frac{4}{3}.$$

Which Method Should I Use?

Situation	Good first method
Obvious bad rows or columns	Dominated strategy elimination
Zero-sum table, pure solution possible	Saddle point test
Two rows or two columns, no saddle point	Graphical mixed strategy method
Larger zero-sum game	LP formulation
Business competition with continuous decisions	Best-response functions

Common Mistakes

- Comparing Player A's payoff with Player B's payoff in the same cell.
- Forgetting that Player B minimizes A's payoff in a zero-sum table.
- Declaring a Nash equilibrium without checking unilateral deviations.
- In graphical method, using the wrong envelope: lower envelope for A's two-row problem, upper envelope for B's two-column problem.
- Forgetting probabilities must sum to one and be nonnegative.

Mini Quiz

- ① In a Nash equilibrium, can the outcome be inefficient?
- ② In a zero-sum game, what does Player B want to do to Player A's payoff?
- ③ What equality must hold for a saddle point?
- ④ If Player A mixes between two rows, what must be true about the expected payoff of those two rows?

Summary

- Game theory studies decisions where payoffs depend on other decision makers.
- Nash equilibrium is a no-unilateral-deviation condition.
- Dominance and saddle point tests are fast first checks.
- Mixed strategies use probabilities to create stable expected payoffs.
- Graphical and LP methods are practical tools for zero-sum games.

For homework

Start each problem by identifying the game type and the solution method before doing algebra.

References Used in Preparation

- IEDA 1250 Lecture 4: Introduction to Game Theory, Prof. Bo Yang, Summer 2026.
- IEDA 1250 Game Theory problem set and lecture exercises, Summer 2026.
- UCSD undergraduate game theory slides, V. Crawford.
- Bonanno, *Game Theory*, 3rd edition.
- Brown CS lecture slides on introductory game theory.